

SAT UZBEKISTAN

Telegram @satuzbhunters

The function g is defined by $g(x) = \frac{5}{9}x - \frac{7}{9}$. What is the value of $g(18)$?

(A) $\frac{38}{9}$

(B) $\frac{52}{9}$

(C) $\frac{83}{9}$

(D) $\frac{97}{9}$

Question 2

$$f(x) = \frac{x+9}{3}$$

For the function f defined by the equation shown, what is the value of $f(-6)$?

(A) -3

(B) -1

(C) 1

(D) 3

Question 3

x	y
0	15
1	22
2	29

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . What is the y -coordinate of the y -intercept of the line in the xy -plane that represents this relationship?

Answer Preview:

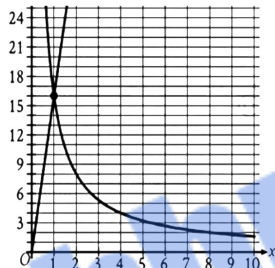
The function f is defined by $f(x) = -9x + 9$. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Answer Preview:

The function f is defined by $f(x) = 2x^2 - 6x$. What is the value of $f(5)$?

Answer Preview:

Question 6



The system of two equations, $y = \frac{16}{x}$ and $y = 16x$, is graphed in the xy -plane as shown. Which ordered pair is a solution to this system?

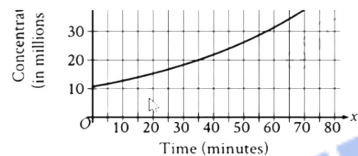
(A) $(0, 0)$

(B) $(0, 16)$

(C) $(1, 16)$

(D) $(16, 16)$

Question 7



The graph shows the estimated concentration of a certain strain of bacteria, in millions of cells per mL of nutrient medium, over time x , in minutes since the initial measurement. According to the graph, which of the following is closest to the number of minutes it took for the estimated concentration of the bacteria to increase from the initial concentration to 20 million cells per mL of nutrient medium?

- (A) 70
- (B) 35
- (C) 10
- (D) 7

Question 8

The function g is defined by $g(x) = -8x + 60$. What is the value of $g(x)$ when $x = -3$?

(A) 84

(B) 49

(C) 36

(D) $\frac{57}{8}$

$$f(x) = x^4 + 6x + 11$$

For the given function f , the graph of $y = f(x)$ in the xy -plane passes through the point $(0, b)$, where b is a constant. What is the value of b ?

Answer Preview:

The function f is defined by $f(x) = x^3 + 17$. What is the value of $f(2)$?

Answer Preview:

Question 11

The function f is defined by $f(x) = 3x - \frac{1}{5}$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

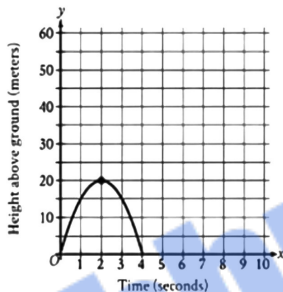
(A) $\left(0, -\frac{1}{3}\right)$

(B) $(0, -3)$

(C) $(0, 3)$

(D) $(0, 5)$

Question 12



An object was launched upward from a platform. The graph shown models the height above ground, y , in meters, of the object x seconds after it was launched. For which of the following intervals of time was the height of the object increasing for the entire interval?

(A) From $x = 0$ to $x = 2$

(B) From $x = 0$ to $x = 4$

(C) From $x = 2$ to $x = 3$

(D) From $x = 3$ to $x = 4$

Question 13

The graph of the polynomial function f in the xy -plane, where $y = f(x)$, passes through the point $(5, 3)$. What is the value of $f(5)$?

(A) 3

(B) 5

(C) 8

(D) 15

Question 14

What is the y -intercept of the graph of $y = 11^x$ in the xy -plane?

(A) $(0, 1)$

(B) $(0, 0)$

(C) $(0, 11)$

(D) $(0, 10)$

What is an x -coordinate of an x -intercept of the graph of $y = 4(x - 13)(x + 5)(x + 1)$ in the xy -plane?

Answer Preview:

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The function f is defined by $f(x) = -12x + 28$. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Answer Preview:

Question 17

In the xy -plane, the graph of the equation $y = 3^x$ is translated 6 units down. What is an equation of the resulting graph?

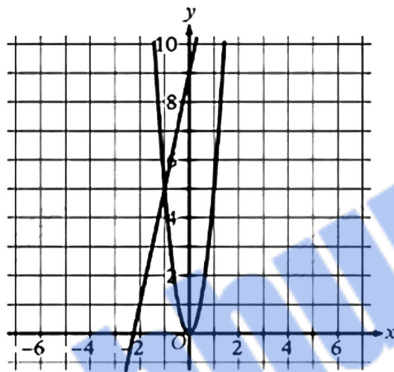
(A) $y = -6(3)^x$

(B) $y = 3^{(x+6)}$

(C) $y = 3^x - 6$

(D) $y = 3^{(x-6)}$

Question 18



The graph of a system of a linear equation and a quadratic equation is shown. Which of the following is a solution (x, y) to the system?

(A) $(-1, 5)$

(B) $(0, 0)$

(C) $(-2, 0)$

(D) $(1, 5)$

Question 19

The function f is defined by $f(x) = 6x - \frac{1}{8}$. What is the y -intercept of the graph of $y = f(x)$ in the xy -plane?

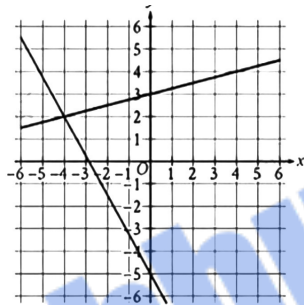
(A) $\left(0, -\frac{1}{8}\right)$

(B) $(0, -6)$

(C) $(0, 6)$

(D) $(0, 8)$

Question 20



The graph of a system of two linear equations is shown. The solution to the system is (x, y) . What is the value of y ?

(A) -5

(B) -4

(C) 2

(D) 3

The function f is defined by $f(x) = 50x$. For what value of x is $f(x) = 150$?

Answer Preview:

$$h(x) = x + b$$

For the linear function h , b is a constant and $h(0) = 21$. What is the value of b ?

Answer Preview:

The function f is defined by $f(x) = \frac{7}{10}x + 63$. What is the value of $f(20)$?

Answer Preview:

$$f(x) = 22$$

For the given linear function f , which table gives three values of x and their corresponding values of $f(x)$?

(A)

x	$f(x)$
0	0
1	0
2	0

(B)

x	$f(x)$
0	22
1	22
2	22

(C)

x	$f(x)$
0	0
1	22
2	44

Question 25

The function f is defined by $f(x) = -3x + 40$. What is the value of $f(x)$ when $x = -6$?

(A) 58

(B) 37

(C) 34

(D) 31

The y -intercept of the graph of $y = x^2 + 15$ in the xy -plane is $(0, y)$. What is the value of y ?

Answer Preview:

Question 27

What is the y -intercept of the graph of $y = 2x^2 + 6x + 3$ in the xy -plane?

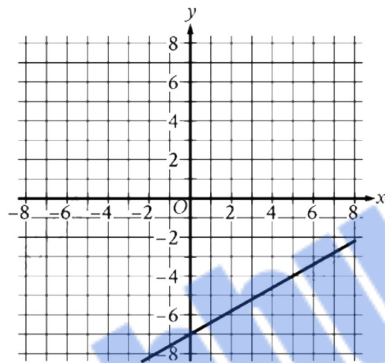
(A) $(0, 2)$

(B) $(0, 3)$

(C) $(0, 6)$

(D) $(0, 9)$

Question 28



The graph of the linear function f is shown, where $y = f(x)$. What is the y-intercept of the graph of f ?

(A) $(0, 7)$

(B) $(0, 0)$

(C) $(0, -5)$

(D) $(0, -7)$

The function f is defined by $f(x) = 7x^3 + 5$. What is the value of $f(2)$?

Answer Preview:

Question 30

What is the y -intercept of the graph of $3x + 2y = 96$ in the xy -plane?

(A) $(0, 5)$

(B) $(0, 6)$

(C) $(0, 32)$

(D) $(0, 48)$

Question 31

The function f is defined by $f(x) = 10x^2 - 38x - 150$. What is the value of $f(0)$?

(A) -150

(B) -38

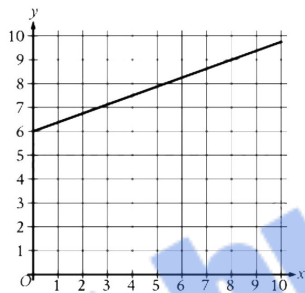
(C) 0

(D) 10

The function g is defined by $g(x) = 2x - 148$. For what value of x does $g(x) = 76$?

Answer Preview:

Question 33



What is the y -intercept of the line graphed?

(A) $(0, -6)$

(B) $\left(0, -\frac{1}{6}\right)$

(C) $(0, 0)$

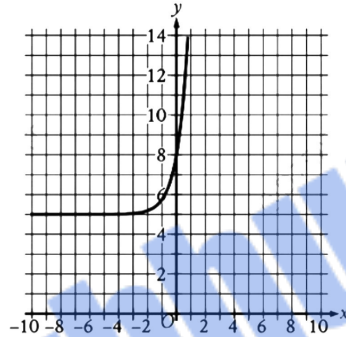
(D) $(0, 6)$

$$p(x) = x^3 + 4x^2 + 5x + 18$$

When the given function p is graphed in the xy -plane, where $y = p(x)$, the y -intercept of the graph is $(0, y)$. What is the value of y ?

Answer Preview:

Question 35



What is the y-intercept of the graph shown?

(A) $(-8, 0)$

(B) $(-5, 0)$

(C) $(0, 5)$

(D) $(0, 8)$

Question 36

$$(v - 12)(v - 3) = 0$$

What are all possible solutions to the given equation?

(A) -12 and 3

(B) -12 and 9

(C) -3 and 12

(D) 3 and 12

Question 37

The function f is defined by $f(x) = \frac{1}{2}(x + 10)$. What is the value of $f(8)$?

(A) 36

(B) 20

(C) 18

(D) 9

The function f is defined by $f(x) = 2x^2 - 5x$. What is the value of $f(8)$?

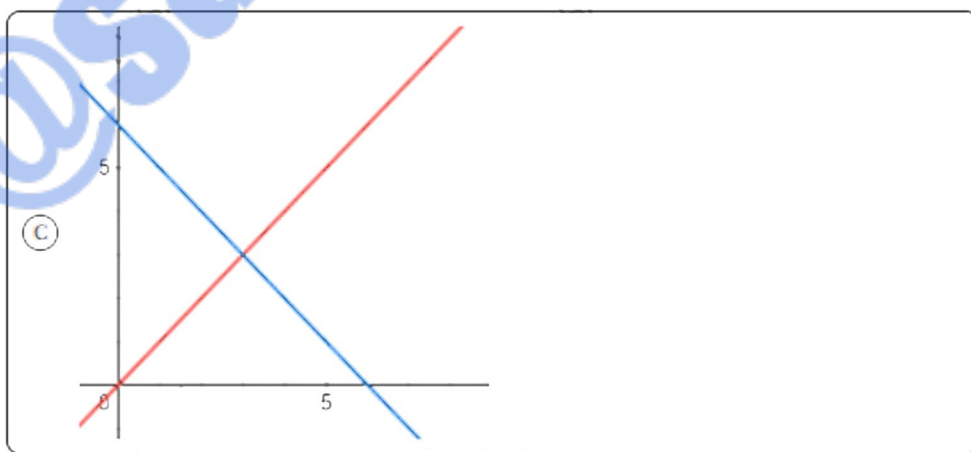
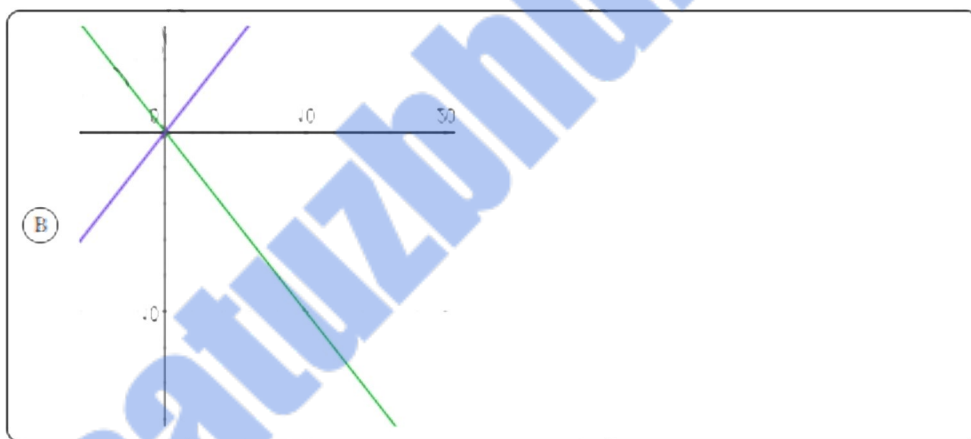
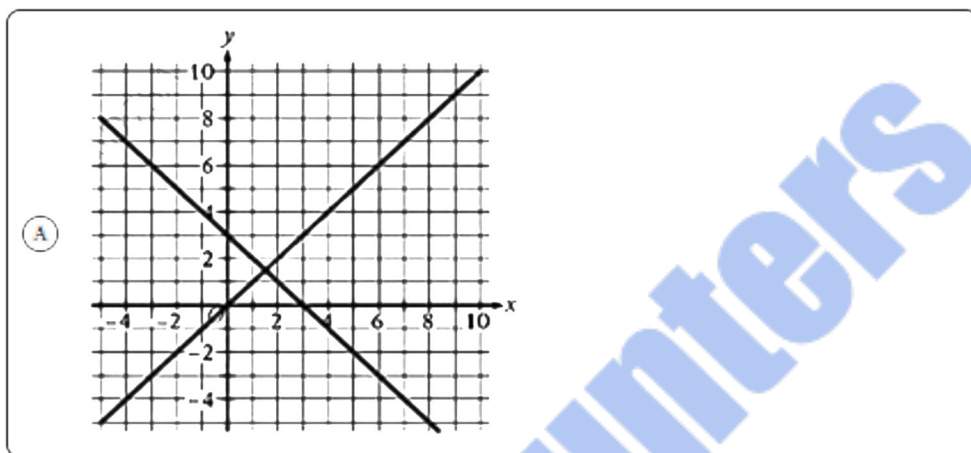
Answer Preview:

If $f(x) = x + 3$ and $g(x) = 3x$, what is the value of $5f(2) - g(2)$

Answer Preview:

Question 40

The solution to a system of linear equations is $(3, 3)$. Which of the following could be the graph of the equations in the system?



Question 41

The graph of the polynomial function f in the xy -plane, where $y = f(x)$, has x -intercepts of $(-9, 0)$ and $(6, 0)$. Which of the following must be true?

(A) $f(0) = -9$

(B) $f(6) = -9$

(C) $f(-9) = 0$

(D) $f(-9) = 6$

Question 42

The function g is defined by $g(x) = -7x + 80$. What is the value of $g(x)$ when $x = -2$?

(A) $\frac{78}{7}$

(B) 66

(C) 71

(D) 94

The function f is defined by $f(x) = 135x - 20$. What is the value of $f(x)$ when $x = 4$?

Answer Preview:

The function f is defined by $f(x) = x^2 + x + 61$. What is the value of $f(2)$?

Answer Preview:

Question 45

The function f is defined by $f(x) = x^2 + 15$. What is the value of $f(2)$?

(A) 2

(B) 15

(C) 17

(D) 19

Question 46

The graph of a polynomial function $y = f(x)$ in the xy -plane passes through the point $(3, 5)$. What is the value of $f(3)$?

(A) 3

(B) 5

(C) 8

(D) 15

Question 47

The graph of the polynomial function f in the xy -plane, where $y = f(x)$, has x -intercepts of $(-5, 0)$ and $(2, 0)$. Which of the following must be true?

(A) $f(0) = -5$

(B) $f(2) = -5$

(C) $f(-5) = 0$

(D) $f(-5) = 2$

Question 48

The function g is defined by $g(x) = 18 - x$. What is the value of $g(3)$?

(A) 3

(B) 15

(C) 18

(D) 21

The function f is defined by $f(x) = 7 + \sqrt{x}$. What is the value of $f(36)$?

Answer Preview:

The function g is defined by $g(x) = 2x - 148$. For what value of x does $g(x) = 78$?

Answer Preview:

x	y
0	12
1	19
2	26

The table shows three values of x and their corresponding values of y . There is a linear relationship between x and y . What is the y -coordinate of the y -intercept of the line in the xy -plane that represents this relationship?

Answer Preview:

Question 52

The function f is defined by $f(x) = 5x^2$. What is the value of $f(6)$?

(A) 30

(B) 40

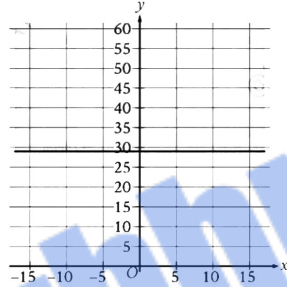
(C) 60

(D) 180

The function f is defined by $f(x) = 12x + 18$. What is the y -coordinate of the y -intercept of the graph of $y = f(x)$ in the xy -plane?

Answer Preview:

Question 54



The shaded region shown in the graph represents all the solutions to which inequality?

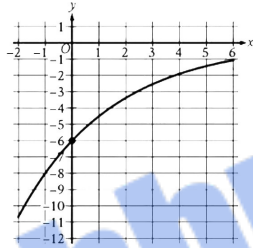
(A) $x \leq 29$

(B) $x \geq 29$

(C) $y \leq 29$

(D) $y \geq 29$

Question 55



The graph of the function f is shown, where $y = f(x)$. What is the value of $f(0)$?

(A) -6

(B) -4

(C) -3

(D) 6

$$p(x) = x^3 + 5x^2 + 8x + 12$$

When the given function p is graphed in the xy -plane, where $y = p(x)$, the y -intercept of the graph is $(0, y)$. What is the value of y ?

Answer Preview:

The function f is defined by $f(x) = x^3 + 14$. What is the value of $f(2)$?

Answer Preview:

The function g is defined by $g(x) = \frac{x}{2}$. For what value of x does $g(x) = 244$?

Answer Preview:

Question 59

The function g is defined by $g(x) = -18 - 9x$. For what value of x does $g(x) = -216$?

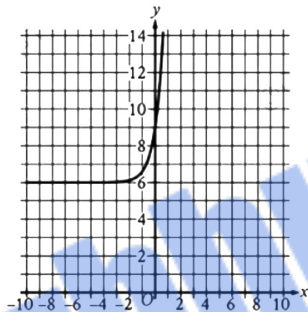
(A) -26

(B) 22

(C) 26

(D) $1,926$

Question 60



What is the y -intercept of the graph shown?

(A) $(-9, 0)$

(B) $(-6, 0)$

(C) $(0, 6)$

(D) $(0, 9)$